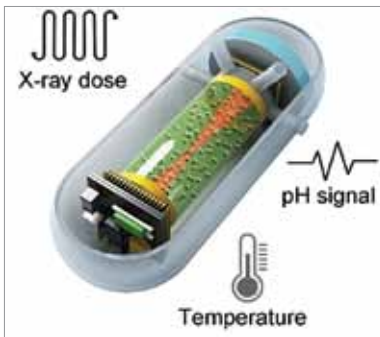




Swallowable X-ray dosimeter estimates radiation dose with neural network based regression model

A joint research team from the Shenzhen Institute of Advanced Technology (SIAT) of the Chinese Academy of Sciences, the National University of Singapore (NUS), and Tsinghua University have developed a 5mm diameter and 0.2mm thickness swallowable X-ray dosimeter that esti-



Functional model of capsule (Credit: SIAT)

mates radiation dose using radioluminescence and temperature with a neural network based regression model. Dosimeters proved 5x more precise than traditional clinical dosimeters, which are placed near the patient's skin.

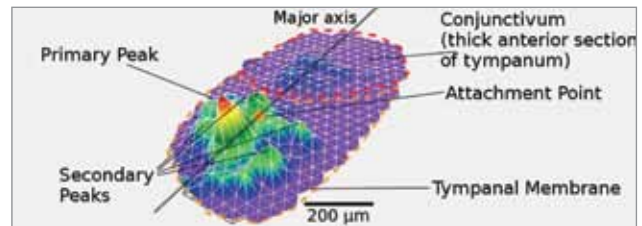
Other in vivo dosimetry methods, such as electronic portal imaging devices, are costly and can modify the patient's dose due to photon attenuation. The capsule dosimeter contains X-ray nanoscintillators, a polyaniline film, a wireless miniaturised luminescence readout system, and can measure pH and temperature. It can measure absorbed dose in gastric cancer radiotherapy and monitor other malignancies with size optimisation.

New method improves wireless device identification and protects inter-device communication

Computer engineers at the University of Illinois Chicago have created a method inspired by quantum physics to improve wireless device identification and protect device-to-device communication. The team created an algorithm to detect exceptional points in RFID systems that use memory chips vulnerable to attack. Unique fingerprints of each device from fabrication variations make it more secure. The device's cryptography structure produces a distinct signal, similar to a person's voice. After testing, the digital fingerprints produced using the exceptional point algorithm passed randomness and security tests. The researchers created a circuit that improves the electromagnetic physically unclonable function's (PUF's) uniqueness and robustness. The lightweight analogue PUF may have various security and anti-counterfeiting applications in wireless communications. The team used standard PCB fabrication to produce a low-cost wireless identification system that utilises the exceptional point.

Bio-inspired microphones gather acoustic data autonomously with low power usage

Researchers at the University of Strathclyde in the UK have developed bio-inspired microphones that can gather acoustic data autonomously with low power usage. The team uses insect ears as inspiration, 3D prints custom materials to mimic insect membranes, which act as sensitive acoustic sensors. 3D printing is vital for flexible,



The displacement of the wax moth *Acroia grisella* membrane, which is one of the key sources of inspiration for designing miniature, bio-inspired microphones (Credit: Andrew Reid)

custom bio-inspired microphones, unlike inflexible silicon based methods. Mic's interesting aspects occur at micro/nanoscales; thickness/porosity vary in the former, while compliance/density vary in the latter. These microphones are bio-inspired not only in their material but also in their data collection process. They target a specific signal, unlike conventional microphones that gather a broad range of information. Similar to nerve endings, this efficient process enables the sensor to quickly detect triggers with minimal energy use and without supervision.

Progress in use of sustainable, cost-effective, and environmentally friendly electrolytes in batteries

Researchers at Huazhong University of Science and Technology have made progress in using organic electrode materials (OEMs) in batteries with aqueous electrolytes, which are more sustainable, cost-effective, and environmentally friendly than conventional organic electrolytes in lithium-ion batteries. The team chose Azobenzene for its low cost, high solubility in organic solvents but not in water. Azobenzene's azo group transfers two electrons reversibly, giving it high capacity. Upon discharge, it absorbs two electrons and becomes hydroazobenzene via two protons' rapid, reversible binding. High capacity is due to the reversible transfer of two electrons, according to analyses. Coin and pouch cells with azobenzene OEMs and zinc counter-electrodes had stable capacities over 200 cycles. Azobenzene OEMs can be easily recycled with commercial organic solvents, unlike polymeric OEMs.